Assignment 4

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Econometrics I
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Instructions. This is the first of the short biweekly assignments: two problems, covering the instrumental-variables designs from Session 7 (examiner or "judge" instruments, and shift-share exposure designs). `R` with `fixest` or Python with `pyfixest`; `feols(y ~ 1 | d ~ z)` estimates 2SLS. Turn in a PDF with your results and reasoning, and your code. Derive first, then simulate to check yourself.

1 Loan officers as instruments

A bank receives 4,000 small-business loan applications, each assigned *at random* to one of 40 loan officers. Officer j approves an application if the firm's creditworthiness score clears the officer's personal cutoff c_j:

$$s_i = u_i + v_i, \qquad D_i = \mathbf{1}\{s_i > c_{j(i)}\}, \qquad c_j \sim \mathcal{N}(0.8,\ 0.5^2) \text{ (one draw per officer)},$$

where $u_i \sim \mathcal{N}(0,1)$ is firm quality (seen by the officer, not by you) and $v_i \sim \mathcal{N}(0,1)$ is paperwork noise. Cutoffs are drawn once per officer. The outcome is the firm's log revenue two years later:

$$y_i = 1 + \tau_i D_i + u_i + \varepsilon_i, \qquad \tau_i = 1 - 0.4\, s_i, \qquad \varepsilon_i \sim \mathcal{N}(0, 0.5^2).$$

The loan helps every firm, but helps firms with better scores less (they have other options). The **average treatment effect** is $\mathbb{E}[\tau_i] = 1$.

(a) Before simulating. OLS of y on a binary D is a difference of two group means (Lecture 1's indicator trick): $\mathbb{E}[y \mid D = 1] - \mathbb{E}[y \mid D = 0]$. Substitute the outcome equation and show this equals

$$\underbrace{\mathbb{E}[\tau_i \mid D_i = 1]}_{\text{effect on the treated}} + \underbrace{\mathbb{E}[u_i \mid D_i = 1] - \mathbb{E}[u_i \mid D_i = 0]}_{\text{selection}},$$

the decomposition from Session 5. Then use the approval rule to sign each piece: which firms get approved, what does that say about their u_i, and what does it say about their τ_i relative to the average of 1? Which piece do you expect to dominate? Give the economic story in one sentence.

(b) Simulate the data. Regress y on D. Then compute the two pieces from (a) directly (you know τ_i and u_i) and confirm they add up. Compare each to 1.

(c) Build the instrument: for each application, the assigned officer's approval rate computed *leaving out application i*. Call it z_i ("leniency"). Why leave i out? (One sentence; think about what D_i itself contributes to the officer's rate.)

(d) Estimate the first stage (regress D on z) and the 2SLS estimate of the effect of a loan. Report both. Interpret the first-stage coefficient in words.

(e) State the three conditions this design needs — relevance, exclusion, and *monotonicity* — in the language of loan officers. Which one does random assignment buy you, which one is about how officers behave, and which one is about whether an officer's strictness affects revenue *other than* through the loan?

(f) Your 2SLS estimate should be noticeably below the ATE of 1. Explain who the *compliers* are in this design (which firms' approval actually depends on the officer they drew), use the model to say what their s_i looks like, and hence why the LATE is below the ATE here. Confirm by computing the average of τ_i among simulated firms whose s_i lies between the strictest and most lenient officer cutoffs.

(g) Managerial summary, two sentences: the bank is considering making every officer as strict as its strictest one. Which of your numbers speaks to that policy, and which number would a consultant quoting "the average effect of a loan" be describing?

2 A shift-share instrument for local demand

A retailer wants to know how much local employment growth raises its stores' sales. It has 300 cities and 8 industries. Cities specialize: each city has one *dominant* industry, chosen at random, with pre-period employment share $s_{ck} = 0.5$; the other seven industries split the rest equally (0.5/7 each). Over the study period each industry gets a *national* growth shock $g_k \sim \mathcal{N}(0, 1)$, drawn once per industry. City employment growth and store sales growth are:

$$x_c = \underbrace{\textstyle\sum_k s_{ck}\, g_k}_{B_c} + 0.5\,\eta_c + \nu_c, \qquad\qquad y_c = 0.5\,x_c + \eta_c + \varepsilon_c,$$

where $\eta_c \sim \mathcal{N}(0,1)$ is a local shock (a new highway, a plant closing) that moves both employment and sales, and $\nu_c, \varepsilon_c \sim \mathcal{N}(0, 0.5^2)$. The retailer also has last period's sales growth, $y_c^0 = \eta_c + \xi_c$ with $\xi_c \sim \mathcal{N}(0,1)$: a noisy look at the local shock. The true effect is 0.5.

(a) Simulate. Regress y on x by OLS. Which direction is the bias, and why?

(b) Construct the *shift-share* (Bartik) instrument $B_c = \sum_k s_{ck} g_k$: each city's exposure to the national shocks, weighted by what it was already doing. Estimate 2SLS with B_c as the instrument for x_c. Compare to 0.5.

(c) Two different assumptions make B_c a valid instrument: (i) the *shares* are as good as randomly assigned with respect to η_c (Goldsmith-Pinkham, Sorkin and Swift 2020), or (ii) the *shocks* g_k are as good as randomly assigned, whatever the shares look like (Borusyak, Hull and Jaravel 2022). Which one holds in this simulation, and what in the setup tells you?

(d) A colleague proposes a check: regress the pre-period outcome y_c^0 on B_c. What should you find if the instrument is valid, and what do you find?

(e) Now break it. Suppose industry 1 is booming ($g_1 = 2$ instead of a random draw), and that cities with a good local shock are more likely to be industry-1 cities: make industry 1 the dominant industry with probability $\Phi(\eta_c)$ (the standard normal CDF), otherwise draw the dominant industry at random from the other seven. Redo (b) and (d). What happened to the 2SLS estimate, which of the two assumptions in (c) failed, and did the check in (d) catch it?

(f) Inference, one paragraph: there are 300 cities but only 8 industry shocks. Explain why the usual city-clustered standard errors overstate how much independent variation the instrument has, and what the "number of observations" effectively is in this design. (You do not need to implement a correction; Adão, Kolesár and Morales 2019 do.)